Transport in the circle code along the generating loop adds one to the integer

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 447-450 and its proof environment opens on line 451, the next line; the pairing is by adjacency and the retained proof range 451-460 is the whole environment.

Editorial note. The retained counter values (chapter 8, section 1, theorem offset 1) make the counted lemma print as Lemma 8.1.2, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:transport-s1-code = 8.1.2).

Editorial note. One label the proof cites lies outside the card, thm:transport-compose (book Theorem 2.3.10, whose chapter file is not part of the pinned corpus so the pointer span is the label line of this chapter dump), and is declared in external_references as a non-load-bearing navigation pointer: the retained proof applies it in one rewrite.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: transport in the circle code along the generating loop and its inverse (author label lem:transport-s1-code, book Lemma 8.1.2).

Allowed standard prerequisites (editorial).

- The circle code. The code family assigns the integers to the base point, and its transport along the generating loop is what the lemma computes.. It is the family being transported in.
- Composition rule for transport. Transport along a path equals transport along the image of the path under the family, followed by transport in the family of types.. It is the first rewrite of the computation.
- Computation of the action on the loop and of univalence. The action of the code family on the generating loop is the successor equivalence on the integers, and transport along its image under univalence is the induced map on the integers.. They are the last two rewrites that produce the arithmetic statement.

Source reference. Original theorem 2.3.10: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1894–1894. Book Theorem 2.3.10, the composition

rule for transport, used in the first rewrite of the retained computation. Navigation pointer only: the chapter file that carries it is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

Lemma 8.1.2. $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, x) = x + 1$ *and* $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}, x) = x - 1$.

Proof. For the first equation, we calculate as follows:

$$\begin{aligned}
\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, x) &= \mathsf{transport}^{A \mapsto A}((\mathsf{code}(\mathsf{loop})), x) \\
&\qquad\qquad \text{(by original theorem 2.3.10)} \\
&= \mathsf{transport}^{A \mapsto A}(\mathsf{ua}(\mathsf{succ}), x) \quad \text{(by computation for } \mathsf{rec}_{\mathbb{S}^1}\text{)} \\
&= x + 1. \qquad\qquad \text{(by computation for } \mathsf{ua}\text{)}
\end{aligned}$$

The second equation follows from the first, because $\mathsf{transport}^B(p, -)$ and $\mathsf{transport}^B(p^{-1}, -)$ are always inverses, so $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}, -)$ must be the inverse of succ. □